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Quantitative Analyses of Coupling in Hybrid Zones

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In hybrid zones, whether barrier loci experience selection mostly independently or as a unit depends on the ratio of selection to recombination as captured by the coupling coefficient. Theory predicts a sharper transition between an uncoupled and coupled system when more loci affect hybrid fitness. However, the extent of coupling in hybrid zones has rarely been quantified. Here, we use simulations to characterize the relationship between the coupling coefficient and variance in clines across genetic loci. We then reanalyze 25 hybrid zone data sets and find that cline variances and estimated coupling coefficients form a smooth continuum from high variance and weak coupling to low variance and strong coupling. Our results are consistent with low rates of hybridization and a strong genome-wide barrier to gene flow when the coupling coefficient is much greater than 1, but also suggest that this boundary might be approached gradually and at a near constant rate over time.

Species are often separated by genetic and phenotypic discontinuities that evolved and are maintained by natural selection. Whether such discontinuities generally arise from selection on one, a few, or many genes is largely unknown (Nosil and Schluter 2011; Nosil et al. 2021). For example, theory suggests that speciation with gene flow occurs more readily when selection is concentrated on a few large effect genes (Yeaman and Whitlock 2011), and there is some empirical evidence consistent with this prediction (Kozak et al. 2019; Unbehend et al. 2021). However, empirical studies also show that speciation can involve divergence at many genes (Michel et al. 2010; Martin et al. 2019; Kautt et al. 2020), although this does not necessarily imply that many genes causally drive speciation. When multiple genes contribute to speciation, linkage disequilibrium (LD) among these genes and other regions of the genome increases the total selection experienced by each locus (Barton 1983; Barton and de Cara 2009). In essence, LD causes selection on causal loci (i.e., barrier loci) to spill over leading to indirect selection at other loci. Barton (1983)

Editors: Catherine L. Peichel, Daniel I. Bolnick, Åke Brännström, Ulf Dieckmann, and Rebecca J. Safran
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referred to this process as "coupling," the usage of which is consistent with the definition of coupling described in perspective 2 of Dopman et al. (2023), but is somewhat different from the definition from Ritchie and Butlin (2023) that emphasizes associations between genes affecting signals and preference. Whether selection affects individual genes or larger genomic regions (e.g., chromosomes or the entire genome) depends on coupling and this in turn influences whether genes or genomes diverge during speciation (Barton and Bengtsson 1986; Wu 2001). The transition to a coupled system where LD causes a genome-wide barrier to gene flow might be important for completing the speciation process (Barton and de Cara 2009; Nosil et al. 2021).

Hybrid zones are powerful systems for studying speciation in general (Peñalba et al. 2023), and specifically for assessing whether selection occurs on barrier loci independently or as a unit creating a genome-wide barrier to gene flow (Abbott et al. 2013; Harrison and Larson 2014; Gompert et al. 2017). In hybrid zones, spatial structure and selection (s) against hybrids maintains LD among barrier loci, whereas recombination (r) in hybrids breaks down these associations. The balance between these processes determines the extent to which barrier loci operate independently or as a unit and this is captured by Barton's coupling coefficient $\theta = s/r$ (Barton 1983). Specifically, Barton (1983) considered a model of simple underdominance, such that an individual heterozygous for n barrier loci has a fitness of $(1-s)^n$; r denotes the average recombination rate between neighboring underdominant loci, such that the total map length with L loci is $R = (L-1)/r$. When θ is greater than 1, the system is coupled and the total selection experienced by each locus (denoted s^*) approaches sL (at least up to $sL = 1$, at which point hybrid fitness is 0), that is, the combined effects of all barrier loci (Fig. 1A; Barton 1983). Whenever coupling occurs, it is s^* (direct and indirect selection combined) rather than s (direct selection due to the causal effect of a locus on fitness) that determines the shape of a cline, particularly the slope near the center of the cline (Barton 1983).

Barton (1983) noted a sharp transition between uncoupled and coupled systems in analytical models at $\theta = 1$ when holding R and the total selection constant but increasing the number of barrier loci L. More recently, results from individual-based simulations have documented a similarly sharp transition in time when populations diverge with gene flow (Flaxman et al. 2014; Nosil et al. 2017). Specifically, divergence remains low until a sufficient number of adaptive mutations (i.e., barrier loci) build-up at which point the system enters a positive feedback loop where divergent selection increases genetic differentiation. This increases LD for the system as a whole, in turn further increasing the selection experienced by each locus. These sharp transitions in space and time suggest that LD among barrier loci can lead to rapid transitions from early to late stages of speciation. However, other theoretical results suggest a smoother transition from uncoupled to coupled systems (i.e., Kruuk et al. 1999) and provide limited evidence for the critical transition in terms of LD among barrier loci at $\theta = 1$ perhaps because of a slow approach to equilibrium expectations (Baird 1995).

Recent work has placed an increasing emphasis on coupling (Bierne et al. 2011; Abbott et al. 2013; Kunerth et al. 2022), and several hybrid zone studies have tested for or analyzed coupling (Vines et al. 2016; Ryan et al. 2017; Cruzan et al. 2021). Nonetheless, we know relatively little about coupling in most hybrid zones. One reason for our limited knowledge is that coupling is not directly measured in the most commonly estimated cline summaries, such as cline concordance and coincidence, and these estimates provide only a binary classification of clines. We aim to overcome these limitations by first using simulations to determine the relationship between the coupling coefficient (θ) and the variance in clines across loci. Specifically, we test the predictions that (1) cline variances decline with increased values of θ, and (2) that the decline in cline variances is nonlinear with a notable transition to lower variances at $\theta = 1$. We then reanalyze genomic data from 25 hybrid zones to infer cline variances and convert these into quantitative estimates of coupling. This allows us to ask whether the hybrid zones we consider exhibit only strong or weak coupling, as predicted by tipping point models, or form more of a connected continuum with a notable zone of intermediate

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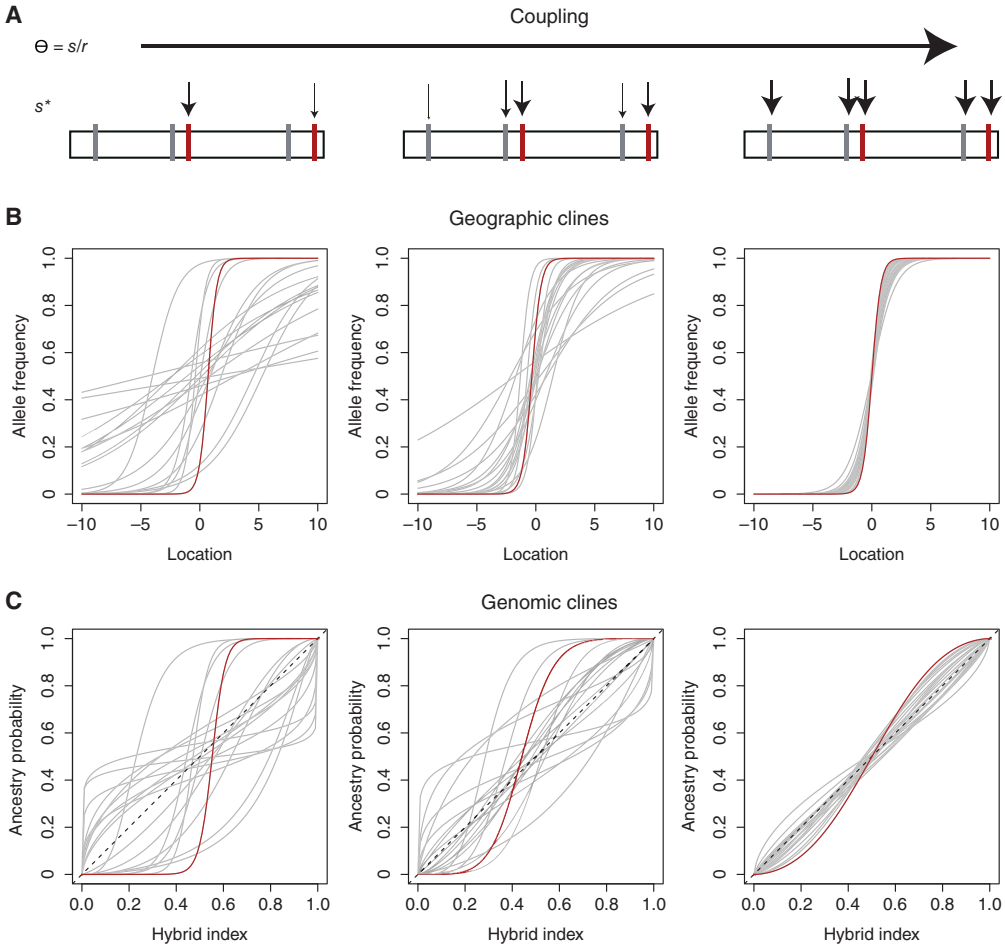


Figure 1. Coupling and its consequences. (*A*) The coupling coefficient $\theta = s/r$ determines the extent to which barrier loci operate independently or as a unit, depicted here with a few example loci. As coupling (θ) increases the total selection (s^*, represented by the size of the vertical arrows) experienced by barrier loci (red bars) and neutral loci (gray bars) increases because of increased linkage disequilibrium. With low coupling, geographic (*B*) and genomic (*C*) clines vary across the genome, whereas with high coupling geographic clines steepen and exhibit similar cline centers and widths and genomic clines converge to the genome-average admixture gradient (hybrid index). Geographic and genomic clines at barrier loci and neutral loci are shown in red and gray, respectively, in *B* and *C*. Dashed lines in *C* denote the genome-average admixture gradient.

systems (i.e., a gray zone of speciation; Roux et al. 2016). We conclude by assessing the consistency of coupling across replicate hybrid zone transects and differential coupling of autosomes versus sex chromosomes based on a subset of these data sets.

MEASURING CLINE VARIANCE

We used two complementary approaches to quantify the variance in clines across a set of loci and evaluate whether the variances relate to the coupling coefficient (θ) in a linear or nonlinear manner (Fig. 1). First, we used geographic clines in allele frequencies. One difficulty with the geographic approach is that expected cline shape differs for cases with and without coupling. Uncoupled, single locus clines are expected to be sigmoidal, whereas coupled, multilocus clines can be better approximated by a steeper sigmoidal function in the center of the cline and

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shallower exponential decay in the tails (Barton 1983; Szymura and Barton 1986). This is because the shape of the central portion of the cline reflects the total selection (s^*) on a locus, including that caused by LD with barrier loci, whereas the shape of the tails of the cline should mostly reflect direct selection on the locus (or on tightly linked loci in LD outside of the hybrid zone). Thus, to provide a common framework for quantifying cline variance using a single model, we focus on the center of each cline—the region governed by total selection (s^*), which is equivalent to s in the uncoupled, single locus model. We fit a linear model for the logit allele frequency, $\log(p/(1-p))$. The slope in this model is expected to be four times the allele frequency gradient (i.e., the reciprocal of cline width) at the center of the cline (i.e., where $p = 0.5$) (Barton and Hewitt 1989; Kruuk et al. 1999). The variance (or rather the standard deviation [SD]) in slope among loci then serves as our metric of cline variance in this model.

Our second approach involved fitting clines in the probability of locus-specific ancestry along a genome-wide admixture gradient (i.e., as a function of a hybrid index; Szymura and Barton 1986; Gompert and Buerkle 2009, 2011). Here, we use the logit–logistic genomic cline function proposed by Fitzpatrick (2013), where the probability a gene copy at locus i for individual j was inherited from species 2 is $\phi_{ij} = (h_j^{v_i})/(h_j^{v_i} + (1 - h_j^{v_i}) * e^{u_i})$. In this equation, h_j denotes the hybrid index (proportion of the genome inherited from a species 2), v_i measures the cline slope (gradient) for locus i relative to the average ($v = 1$) and u_i defines the cline center for locus i relative to the genome average and to v_i (Fitzpatrick 2013). Consequently, information on ancestry at a locus depends on the genotype, an individual's hybrid index (based on all loci), and the genomic cline parameters (based on all of the analyzed hybrids) (Gompert and Buerkle 2011). We use a reparameterization following Bailey (2022) with $\text{logit}(c_i) = u_i/v_i$ to define the more intuitive cline center parameter (c_i) that indicates the hybrid index value at which $\phi_{ij} = 0.5$ (i.e., the probability of ancestry from both species is equal). We measure the variance in clines here as the SD in $\log(v_i)$ and $\text{logit}(c_i)$, both of which have expected means of 0 (i.e., the values averaged across all loci used to define the hybrid index should be 0).

Each of these models has benefits and drawbacks. Geographic clines can only capture hybrid zone dynamics when there is a geographic gradient of gene flow and the scale of cline parameters are dependent on the geographic scale of dispersal, which varies among organisms and is often poorly known. Genomic clines are always relative to genome-average admixture (hybrid index) and thus are not in geographic units, but instead measure the change in the ancestry probability per unit change in hybrid index. This makes the variance in genomic clines easier to compare across hybrid zones, however genomic clines cannot capture the absolute extent of introgression and fail when hybrids are rare (i.e., when the hybrid zone lacks an admixture gradient in h). For this reason, we consider both geographic and genomic clines in simulations (where the scale of dispersal is set), but consider only genomic clines in our analyses of empirical data sets.

We take a hierarchical Bayesian approach to fitting geographic and genomic clines. This is important because it allows us to explicitly estimate the cline variances as parameters in our models. Thus, we assume that the slopes (β) for geographic clines and $\log(v)$ and $\text{logit}(c)$ for genomic clines at each locus represent independent draws from a higher level (normal) distribution with a mean and SD. We are most interested in the SD that describes the variability in clines across the genome. For the geographic cline analysis, we estimate both the mean and SD. We placed a weakly informative normal prior (mean = 0, SD = 5) on the mean (μ_β) and a weakly informative gamma prior (shape = 0.1, rate = 0.01) on the SD (σ_β). In the case of the genomic clines, we expect means of 0 and set them as such (i.e., we use soft centering but do not impose a sum-to-zero constraint), but we estimate the SDs (σ_v and σ_c) by setting weakly informative normal priors (mean = 0, SD = 1) on both. We fit these models using Hamiltonian Monte Carlo (HMC), which is an algorithm that allows for efficient exploration of and sampling from complex posterior probability distributions (Neal

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2011). This allows us to better estimate these SD parameters, which can exhibit poorer mixing using alternative Markov chain Monte Carlo methods (Z Gompert, pers. observation). Model fitting was done using Stan via the R (versions 4.1 and 4.2) interface, *rstan* (version 2.21.7).

SIMULATIONS CONNECTING CLINE VARIANCE TO COUPLING

We simulated hybrid zones with known coupling coefficients and analyzed the simulated data using the models described in the preceding section to measure the association between coupling and the variance in cline parameters. We were especially interested in whether there was a sharp decrease in cline variance at a coupling coefficient of $\theta = 1$. Hybrid zones were simulated using the *dfuse* software described by Lindtke and Buerkle (2015). This software runs individual-based simulations of secondary contact using a stepping stone model and tracks ancestry junctions (Baird 1995; Buerkle and Rieseberg 2008). We modified the existing software to include a model of multiplicative underdominance equivalent to that considered by Barton (1983), that is, where the fitness of a hybrid heterozygous at n barrier loci is $(1-s)^n$. All simulations included 110 demes, each with an adult carrying capacity of 50, arrayed in a one-dimensional stepping-stone model. We set the migration rate between neighboring demes to either 0.1 or 0.2. This set of conditions was chosen to approximate a pair of hybridizing species distributed continuously in space using a large number of small, well-connected demes. We simulated hermaphroditic, diploid organisms with a single 1 Morgan chromosome (thus, there was one expected recombination per meiosis). The simulations assumed random mating within demes with viability selection on progeny. Each generation began with a reproduction phase, which involved creating offspring until either the progeny carrying capacity was reached (100 individuals or twice the adult carrying capacity) or until all of maternal gametes from the adults from the previous generation were exhausted (five per individual). This was followed by progeny dispersal and mortality selection where the probability of survival was given by hybrid fitness, that is $(1-s)^n$. Surviving progeny (now adults) were then culled randomly to the adult carrying capacity if more than 50 progeny survived (see Lindtke and Buerkle 2015 for a full description of the simulation model).

We conducted simulations with coupling coefficients, $\theta = 0.05, 0.1, 0.3, 0.5, 0.7, 0.9, 1, 1.1, 1.5$, and 2, and with the number of underdominant barrier loci set to $L = 2, 10, 100, 200, 500$, or 1000. These were spaced equally across the chromosome, with an average recombination rate between neighboring loci of $r = 1/(L-1)$. We then used the relationship $\theta = s/r$ to calculate the appropriate per locus selection coefficient to achieve the desired coupling coefficient (a few combinations of L and θ would have required $s > 1$ and fitness < 0; these combinations were dropped). We ran the simulations for 2000 generations but examined output every 500 generations to verify that the clines appeared stable (a lack of change could reflect an approximate equilibrium outcome or very gradual approach to a not-yet-reached equilibrium; see Baird 1995). In total, 1140 hybrid zone data sets were simulated. In each case, genotypes at 51 diagnostic (fixed differences between species) biallelic markers spaced evenly along the chromosome (every 2 cM) were output for analysis.

To run the geographic cline analyses, we defined the center of each simulated hybrid zone as the deme where the mean allele frequency was closest to 0.5 (demes near the outer edges of the simulated hybrid zone [i.e., demes 1 to 20 and 90 to 110] were excluded from consideration). We then focused our analyses on the 11 demes centered on this central deme (i.e., the central deme plus five on each side). We fit our hierarchical Bayesian geographic cline models to the logit allele frequencies from these 11 sites. Our focus on 11 sites represents a compromise between avoiding the exponential decay portion of multilocus clines, retaining a sufficient number of demes for analysis, and not unduly constraining the possible values for the slope. This differs from the threshold of logit(p) between -2 and 2 used by Kruuk et al. (1999), which would not have been as practical across the range of simulated hybrid zones. We fit each cline model using

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HMC with 20 chains each comprising 1200 iterations and a 1000 iteration warmup. We computed the Gelman–Rubin convergence diagnostic to verify adequate HMC mixing and likely convergence of the HMC algorithm to the posterior distribution.

Across the 1140 simulated hybrid zone data sets, the mean cline slope (μ_β, measured in units of change in logit allele frequency per deme) ranged from −1.65 to −0.11 (mean = −0.88) and the SD in slopes (σ_β) ranged from 0.0027 to 0.67 (mean 0.26). Together, the mean and SD in cline slopes explained 85.2% of the variation in the simulated coupling coefficients (standardized regression coefficients from simple linear regression with an interaction term: $|\mu_\beta| = 0.47$, s.e. = 0.008; $\sigma_\beta = -0.027$, s.e. = 0.008; $|\mu_\beta|{:}\sigma_\beta = -0.084$, s.e. = 0.010; model P-value < 0.0001) (Fig. 2A). Because the scale of variability in clines (σ_β) depends on the average slope, we also considered the coefficient of variation (CV), that is, $\sigma_\beta/|\mu_\beta|$. We found that the CV was negatively associated with θ and by itself explained 74.0% of the variation in this parameter for the simulated data sets. Importantly, we found that μ_β and the CV varied smoothly with θ, without an obvious, abrupt transition in these high-level cline parameters at $\theta = 1$. With that said, there was evidence that μ_β and CV approached an asymptote around $\theta = 1$ (Fig. 2B,D; supplemental Table S1). A similarly nonlinear relationship was detected between σ_β and θ, such that the SD in clines increased and then decreased as a function of the coupling coefficient with the transition occurring around $\theta = 1$ (Fig. 2C; supplemental Table S1). This transition was especially pronounced when viewed on a log–log scale, where there was evidence of a bimodal distribution of σ_β and possible bistability around $\theta = 1$ as is sometimes observed at tipping points (supplemental Fig. S1; e.g., Nosil et al. 2017).

To fit genomic clines, we first designated the subset of demes with mean allele frequencies between 0.1 and 0.9 to constitute the hybrid zone (as opposed to nonhybrid parental populations). Such demes need not be contiguous and do not necessarily include only or even mostly hybrid individuals, but intermediate allele frequencies at the diagnostics marker loci suggest at least the possibility for hybrids. We then sampled up to 300 individuals as putative hybrids from these demes. We only fit models when one or more demes met this criterion. We used the known hybrid indexes and parental allele frequencies from *dfuse* and fit the hierarchical Bayesian genomic cline models described above using HMC. We ran four independent HMC chains per data set with 2000 total iterations and a 1000 iteration burn-in for each chain. As with the geographic cline analysis, we used the Gelman–Rubin convergence diagnostic to verify good HMC performance.

The total number of hybrids, defined here as individuals with hybrid indexes between 0.1 and 0.9, in the simulated hybrid zones declined with increasing coupling coefficients (linear regression $r^2 = 0.71$, $P < 0.0001$) (supplemental Fig. S2). The SDs for genomic cline center (σ_c) and slope (σ_v) together explained 52.3% of the variation in the simulated coupling coefficients (standardized regression coefficients from linear regression: $\sigma_c = -0.0167$, s.e. = 0.013; $\sigma_v = -0.24$, s.e. = 0.013; $\sigma_c{:}\sigma_v = 0.097$, s.e. = 0.0090; model P-value < 0.0001) (Fig. 3A). These results consider only simulated data sets with at least 10 hybrids (defined as individuals with hybrid indexes between 0.1 and 0.9); similar results were obtained when analyzing data sets that included 50 or more hybrids (see supplemental Fig. S3). Using polynomial regression, we found some evidence of a nonlinear relationship between the coupling coefficient (θ) and these cline hyperparameters (σ_c and σ_v) (Fig. 3; supplemental Table S2), but we did not detect a sharp transition for either parameter around $\theta = 1$. However, we were only able to analyze a modest proportion of data sets with $\theta > 1$ (e.g., 34% with $\theta = 1.5$ and 2% with $\theta = 2$, compared to 83% with $\theta = 1$ and all simulations with $\theta \leq 0.5$) as these produced few actual hybrids, often too few to have a reasonable hybrid index axis for cline fitting (see supplemental Fig. S2). Thus, the lack of hybrids when coupling coefficients notably exceed 1 makes detecting such a sharp transition difficult but at the same time represents a transition to strong reproductive isolation. In other words, while we cannot estimate the variance in clines without hybrids, we can conceptualize such hybrid zones as having a cline variance of 0.

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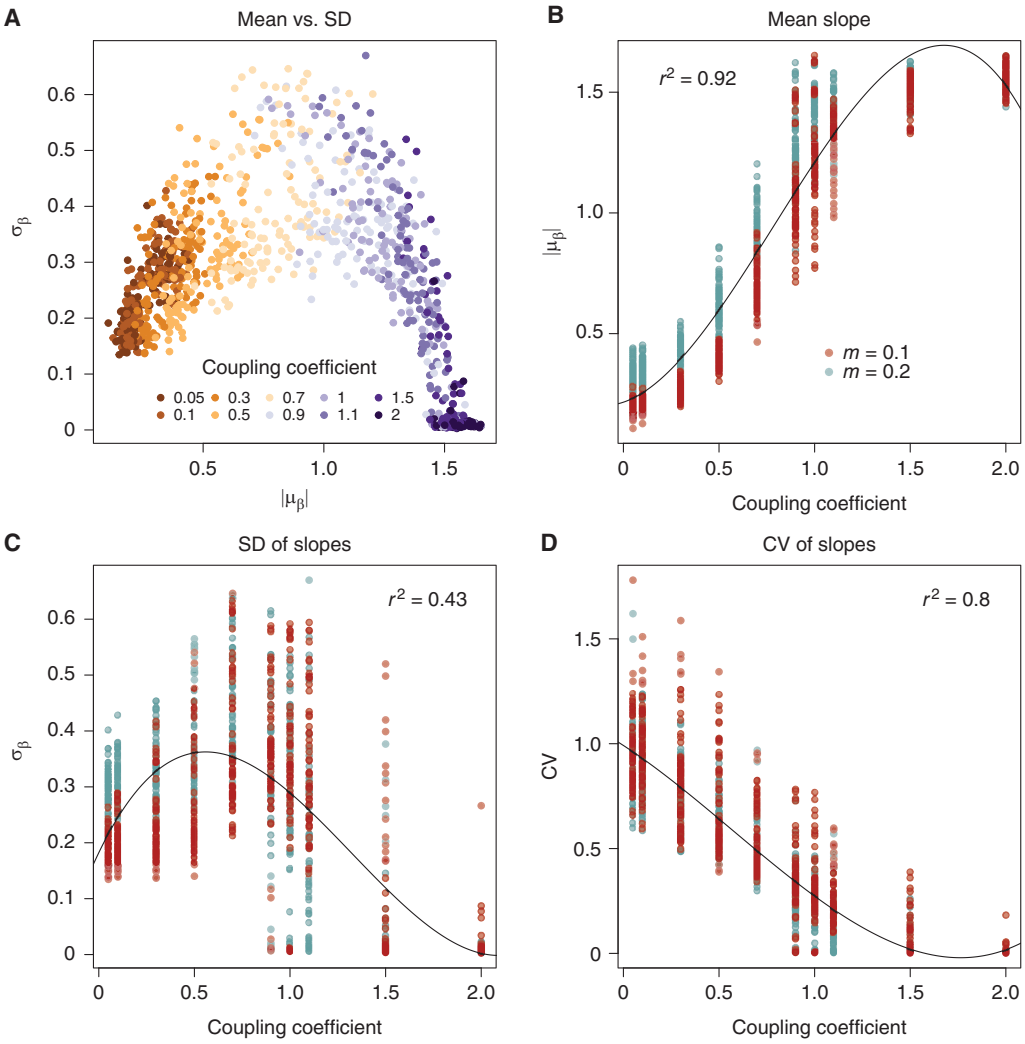


Figure 2. Relationship between geographic clines and coupling in simulated hybrid zones. Panel *A* shows estimates of mean cline slope (μ_β) and the standard deviation (SD) in slopes (σ_β) from the simulated hybrid zones. Points are colored by the known coupling coefficient. Panels *B–D* show the relationship between the coupling coefficient (θ) and μ_β (*B*), σ_β (*C*), and the coefficient of variation (CV) for cline slopes (*D*). Points denote results from individual simulations and are colored based on the migration rate (*m*) between neighboring demes. The best fit line from polynomial regression is shown along with the corresponding coefficient of determination (r^2).

Overall, our geographic and genomic analyses of simulated hybrid zones revealed a strong relationship between the coupling coefficients (θ) and the cline hyperparameters (σ_β and μ_β for geographic clines and σ_c and σ_v for genomic clines), with mixed evidence of a smooth versus more abrupt transition in these parameters as a function of θ. Importantly, even if patterns for clines vary continuously with θ, feedbacks in nature could cause most hybrid zones to fall into a low coupling state with high cline variance (and wide mean geographic clines) or a high coupling state with low cline variance (and steep mean geographic clines). It is to this topic of

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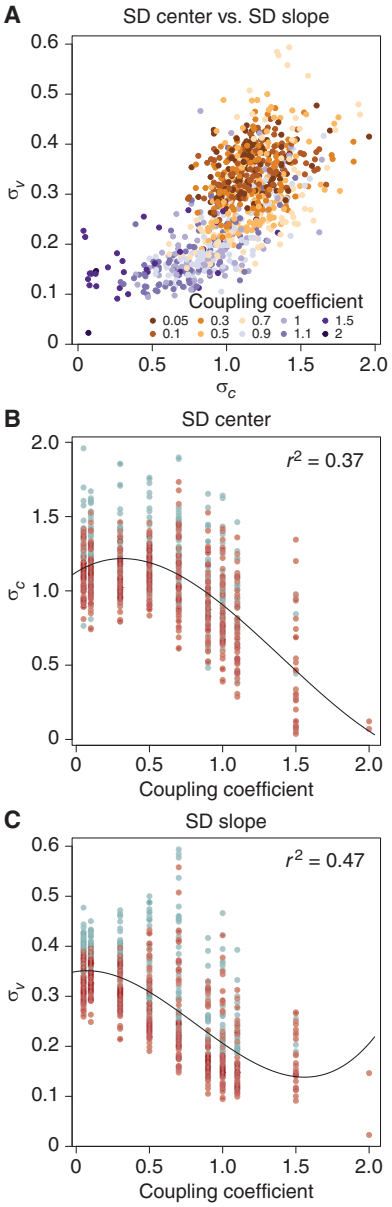


Figure 3. Relationship between genomic clines and coupling in simulated hybrid zones. Panel *A* shows estimates of the standard deviation (SD) for genomic cline center (σ_c) and slope (σ_v) from the simulated hybrid zones. Points are colored by the known coupling coefficient. Panels *B* and *C* show the relationship between the coupling coefficient (θ) and σ_c (*B*) or σ_v (*C*). Points denote results from individual simulations and are colored based on the migration rate (m) between neighboring demes (see Fig. 2). The best fit line from polynomial regression is shown along with the corresponding coefficient of determination (r^2).

natural hybrid zones that we now turn. Although our analyses of simulations suggested a stronger relationship between geographic cline parameters and coupling than between genomic cline parameters and coupling, this was for a set of simulations conducted on a similar scale of dispersal. This will not be true for diverse natural systems making geographic comparisons difficult. We thus focus exclusively on genomic clines for natural hybrid zones. This also allows us to analyze mosaic hybrid zones (e.g., Harrison 1986; Bierne et al. 2003) and other hybrid zones lacking a major geographic axis.

TESTING ROBUSTNESS OF OUR ESTIMATES OF COUPLING FROM SIMULATED DATA

Our goal was to use the relationship between cline parameters and θ documented above to estimate θ in empirical data sets. Before doing so, we conducted several tests of the robustness of our results based on simulations. These are detailed in the online supplemental material, but we highlight the key results here. First, we compared the performance of the linear regression models used to estimate θ to a nonlinear regression approach, specifically the random forest machine learning model. The out-of-bag (i.e., for simulations not used to fit the model) percent variance explained from a random forest model relating σ_c and σ_v to θ was similar to the explanatory power of a linear regression model (random forest out-of-bag variance explained = 58.0% compared to linear model $r^2 = 0.523$), suggesting robustness of our analytical approach. We also found that estimates of the cline parameter SDs were reasonably robust to alternative sampling schemes within the simulated hybrid zones (see the supplemental material; supplemental Fig. S4). Likewise, our results were qualitatively consistent for different genetic map sizes. Specifically, across a set of simulations that included 0.5, 1, or 2 Morgan chromosomes, σ_c and σ_v explained 37% of the variation in the simulated values of θ (linear regression, model $P < 0.0001$) (see the supplemental material; supplemental Fig. S5 for details). Nonetheless, differences in map size quantitatively affected patterns of hybridization and introgression such

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Table 1. Empirical hybrid zone data sets

Organism	Taxonomic group	Divergence time (MYA)	Number of loci	SD cline center (σ_c)	SD cline slope (σ_v)	θ	References
Agalychnis	Amphibian	–	25	1.42	0.56	0.12	Akopyan et al. 2020
Alouatta	Mammal	~3.0	1000	1.02	0.23	0.65	Baiz et al. 2019
Coenonympha	Insect	<0.02	81	1.11	0.31	0.44	Capblancq et al. 2020
Corvus	Bird	~0.443	588	0.86	0.40	0.14	Slager et al. 2020
Crotalus	Reptile	3.0–5.2	1000	0.91	0.33	0.37	Nikolakis et al. 2022
Encelia	Plant	~1.4	1000	0.38	0.31	0.41	DiVittorio et al. 2020
Fundulus	Fish	–	1000	0.45/0.33	0.12/0.08	1.30/1.58[a]	Schaefer et al. 2016
Gryllus	Insect	~0.2	110	0.81/0.48	0.24/0.15	0.67/1.15[a]	Larson et al. 2014
Hirundo	Bird	<0.1	54	0.53	0.31	0.42	Scordato et al. 2017
Iris	Plant	–	1000	1.16	0.19	0.71	Sung et al. 2018
Lissotriton	Amphibian	~1.0	737/730	0.35/0.97	0.14/0.14	1.26/0.92[a]	Zieliński et al. 2019
Lycaeides	Insect	~2.4	500	0.94	0.35	0.31	Chaturvedi et al. 2020
Mus	Mammal	~0.5	1000	0.69/0.70/1.01	0.16/0.14/0.23	1.00/1.07/0.66[a]	Janoušek et al. 2012
Mytilus	Bivalve	~3.5	419	0.29	0.27	0.62	Saarman and Pogson 2015
Nematocharax	Fish	0.27–0.67	52	0.80	0.41	0.08	Barreto et al. 2020
Neotoma	Mammal	~1.6	623	0.52	0.26	0.64	Jahner et al. 2021
Oleria	Insect	–	1000	1.73	0.50	0.42	Gauthier et al. 2020
Papilio	Insect	0.5–0.6	164	0.29	0.21	0.93	Ryan et al. 2017
Papio	Mammal	1.0–2.2	501	0.84	0.41	0.10	Chiou et al. 2021
Picea	Plant	12.5–15.0	221	0.67	0.22	0.77	Hamilton et al. 2013
Pinus	Plant	~18.04	670	1.10	0.37	0.29	Menon et al. 2021)
Poecile	Bird	~1.5	1000	0.94/0.82	0.40/0.38	0.17/0.19[a]	Unpublished
Sceloporus	Reptile	0.045–1.9	38	1.09	0.21	0.69	Leaché et al. 2017
Sternotherus	Reptile	<4.0	798	0.57	0.28	0.55	Scott et al. 2019
Yucca	Plant	0.1–0.2	1000	0.64	0.15	1.06	Royer et al. 2016

(SD) Standard deviation, (Θ) coupling coefficient.
[a]Denotes that there were multiple transects that had coupling coefficients calculated for these organisms.

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that the number of hybrids and the SD in clines declined more rapidly with increased θ when recombination occurred less frequently (supplemental Fig. S5A). Consequently, a regression model incorporating an interaction between map size and the cline SDs increased our ability to explain variation in θ from $r^2 = 0.371$ to $r^2 = 0.534$.

Last, we conducted a full additional set of simulations with a lower rate of gene flow ($m = 0.05$). We found that the relationship between cline parameters and θ was weaker in these simulations (linear model $r^2 = 0.415$, random forest out-of-bag variance explained = 50.6%) (supplemental Fig. S6), and that estimates of coupling coefficients for these simulations based on models fit from the original simulations with $m = 0.1$ or 0.2 were less accurate (estimated values of θ based on the linear regression model fit for $m = 0.1$ and 0.2 explained 22.5% of the variation in the true, simulated values of θ for $m = 0.05$). Importantly, random forest regression models fit for $m = 0.1$ and 0.2 failed to predict θ values for $m = 0.05$ (out-of-bag variance explained ~0%) (in contrast both linear regression and random forest performed well in terms of predicting $m = 0.1$ results from models fit based on $m = 0.2$ and vice versa; see supplemental Fig. S7). Because of this, we focus mostly on the linear regression model for predicting θ with empirical hybrid zones. We also caution that our estimates of θ for the empirical data sets should be taken as estimates given the specific simulation conditions and linear model used here (we discuss ways that this could be improved in the Concluding Remarks section). However, our estimates of cline SDs for the empirical hybrid zones are themselves informative about the variability of clines across the genome and thus the extent that evolutionary processes are operating on all loci similarly, which is central to understanding hybrid zone dynamics and speciation regardless of the connection to a theoretical coupling coefficient.

Empirical Data Sets

We fit genomic cline models for 25 empirical hybrid zones (see Table 1 for details) to determine whether these data sets smoothly spanned the continuum from weak to strong coupling (or likewise, high to low cline variances). We chose these data sets to span taxonomic diversity and only included hybrid zones with genome-wide single nucleotide polymorphism (SNP) data. This is not meant to be an exhaustive set of hybrid zones, but we hope it is representative of the genomic hybrid zone data sets that exist in the literature. Our use of these data sets to estimate θ assumes that each is a hybrid zone maintained by selection against hybrids (e.g., a tension zone) or at least that hybrids are not favored within the hybrid zone (as they would be in models of bounded hybrid superiority) (Moore 1977). Past theory and reviews suggest that most hybrid zones are tension zones or at least exhibit clines similar to expectations from tension zones (Barton and Hewitt 1985). Moreover, by using genomic rather than geographic clines, our analyses can extend to cases where hybridization does not occur in a simple geographic context (e.g., to mosaic hybrid zones). Thus, we expect this assumption to be at least reasonable for most of the data sets, and even if a few do not fit the tension zone model very well, this should not qualitatively alter our core conclusions. Details of the data sets and data processing are provided in the online supplemental material.

For each data set, we analyzed only ancestry informative loci (here defined as SNPs with an allele frequency difference between the parental taxa > 0.3), and we limited our analysis to 1000 randomly sampled loci (mean number of loci = 583, SD = 389, minimum = 25, maximum = 1000). We chose a minimum allele frequency difference cutoff of 0.3 to ensure loci carried sufficient information about genetic ancestry to provide meaningful estimates of genomic clines (clines in ancestry), while also not excluding data sets with lower levels of genetic differentiation (see, e.g., Gompert et al. 2012). Still, this does not ensure an identical level of ancestry informativeness of loci across data sets. We limited the number of loci to 1000 as our goal was to efficiently estimate the variance in clines (even a sample size of 25 provides a relatively precise estimate of a variance) not to describe detailed

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patterns of introgression across the genome. Additionally, using no more than 1000 loci minimizes the effect of LD caused by tight physical linkage on our results (as opposed to LD more generally, which is a signal of interest). Cline variances were inferred using the Bayesian genomic cline model described above with eight HMC chains each comprising a 1000-iteration warmup and 1500 iterations for sampling. We then estimated coupling coefficients based on the values of σ_c and σ_v and the parameterized linear model or random forest regression model from the simulated data sets (the parameterized models were based on the core simulations with $m = 0.1$ or 0.2).

For the 25 empirical hybrid zones, estimated SDs in genomic cline center (σ_c) and slope (σ_v) ranged from 0.29 to 1.73 (mean = 0.83) and 0.12 to 0.56 (mean = 0.29), respectively, and thus broadly overlapped with estimates from our simulated hybrid zones (Figs. 4 and 5; see supplemental Fig. S8 for the distribution of hybrid indexes in each hybrid zone). Using the linear model fit on the simulated data sets, estimates of the coupling coefficient (θ) for the empirical hybrid zone data sets ranged from 0.08 to 1.30 (mean = 0.55). We found no evidence of a gap in inferred coupling coefficients, rather we documented a smooth continuum from low to high coupling across these 25 data sets (Fig. 5). Moreover, similar results were obtained when estimating θ from our fit random forest regression model; this was true both with respect to the distribution of coupling coefficients across the data sets (supplemental Fig. S9) and the specific estimates of θ for each data set (Pearson correlation = 0.82, 95% confidence intervals (CIs) = 0.63–0.92) (supplemental Fig. S10). As noted in the preceding section, both the linear regression and random forest models were parameterized from a specific set of simulations and the relationship between the cline variances and θ could (and likely does) vary under different conditions. However, we also documented a smooth continuum in the cline SDs (σ_c and σ_v) across the 25 hybrid zone data sets. This suggests that even if these metrics do not relate to the theoretical parameter θ in the manner suggested here, we still find a smooth continuum in the degree to which loci introgress independently (uncoupled) or not (coupled) across these hybrid zones, and it is this that is most relevant for our understanding of speciation.

Our analyses above consider a single hybrid zone for each species, but the evolutionary outcomes of secondary contact can vary among populations due to genomic and ecological context (Gompert et al. 2017; Mandeville et al. 2017). Thus, to assess this possibility with respect to coupling and thereby further evaluate the robustness of our general results to details of the systems considered, we compared estimates of coupling across transects for a subset of species. Specifically, the data sets we obtained included five species with two hybrid zones or transects, and one species (*Mus*) with three transects, resulting in eight pairs of transects (only one transect from each was included in the core analyses above). Using these additional transects, we found moderate consistency in estimates of θ for the pairs (Pearson correlation = 0.67, 95% CI = −0.17–0.95, $P = 0.101$, similar results were obtained for estimates based on random forest regression, $r = 0.75$, 95% CI = −0.01–0.96, $P = 0.053$) (supplemental Fig. S10; supplemental Table S3).

Last, given the widespread interest in the role of sex chromosomes in speciation (Haldane 1922; Coyne and Orr 1989) and observation that sex chromosomes often have steeper clines across hybrid zones (e.g., Tucker et al. 1992; Carling and Brumfield 2008; Hooper et al. 2019), we asked whether coupling was stronger (higher estimates of θ) for sex chromosomes than autosomes. For this, we ran additional analyses separately estimating cline SDs and θ for eight hybrid zone data sets (including replicate transects) that we were especially familiar with and where there was clear information about which SNPs were on the X or Z sex chromosome versus autosomes (see supplemental Table S4). Importantly, differences in recombination rates or effective population size for sex chromosomes relative to autosomes, in addition to differences in the number of barrier loci, could contribute to differences in estimates of θ. Across these data sets, estimates of θ for autosomes and sex chromosomes were positively correlated (Pearson correlation = 0.65, 95% CI =

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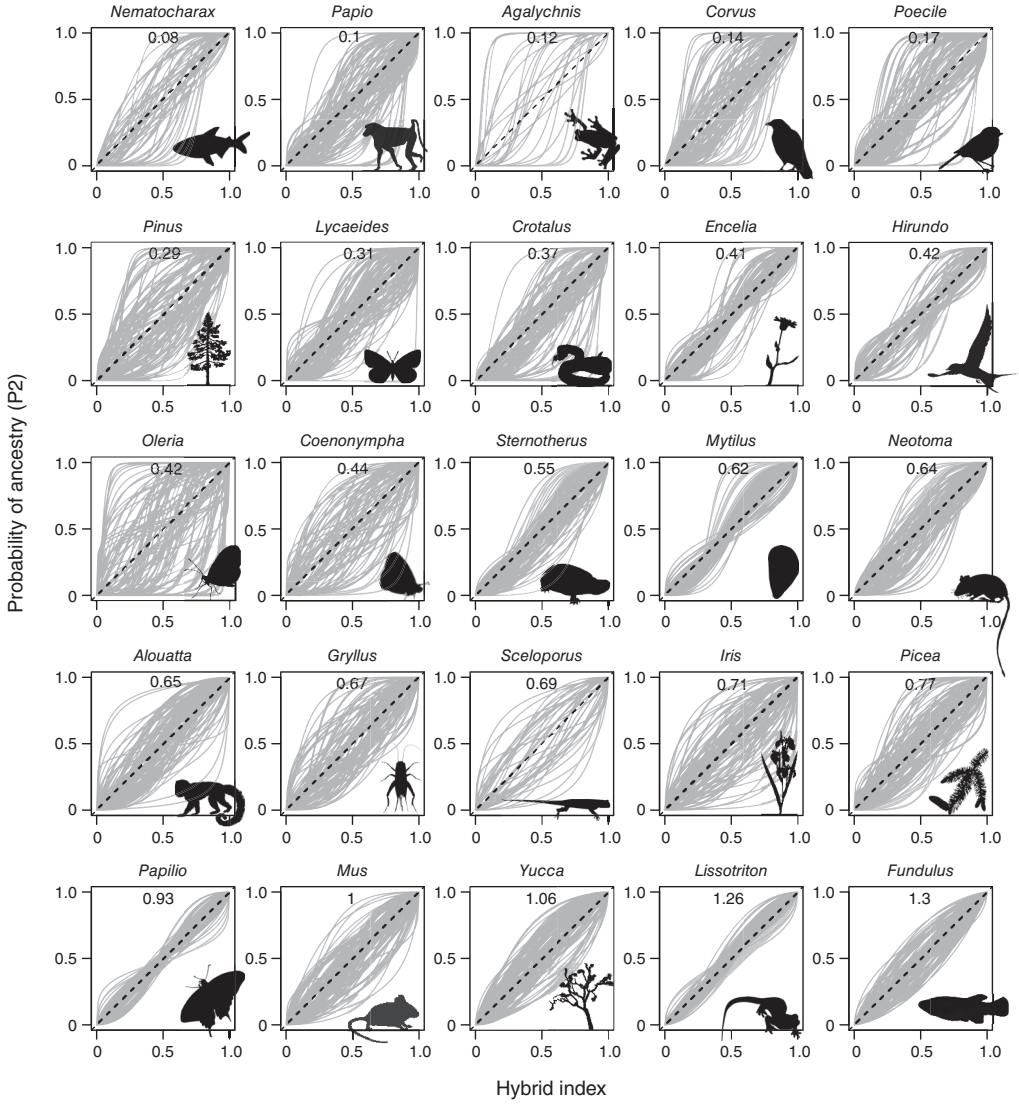


Figure 4. Summary of genomic clines empirical hybrid zones. The plots show estimated genomic clines (gray lines) for each of 25 hybrid zones. Each cline denotes the probability of ancestry for species 2 as a function of hybrid index (the total proportion of the genome inherited from species 2). Clines for 100 randomly chosen loci (or all loci if there were fewer than 100) are shown. The dashed one-to-one line denotes an ancestry probability equal to hybrid index. Our estimate of the coupling coefficient based on the variability among clines is reported.

−0.09–0.93, $P = 0.078$). Moreover, in seven of the eight data sets θ was higher for the X/Z chromosome than for the autosomes (mean difference in θ across data sets = 0.12) (supplemental Table S4). The one exception was for the *Gryllus* Connecticut transect where θ was notably higher for the autosomes (0.86) than for the X chromosome (0.39).

CONCLUDING REMARKS

We documented a smooth continuum of coupling across 25 natural hybrid zones, ranging from very weak coupling to near-complete coupling suggestive of a strong genome-wide barrier to gene flow (θ > 1) (Fig. 5B). As such, we found

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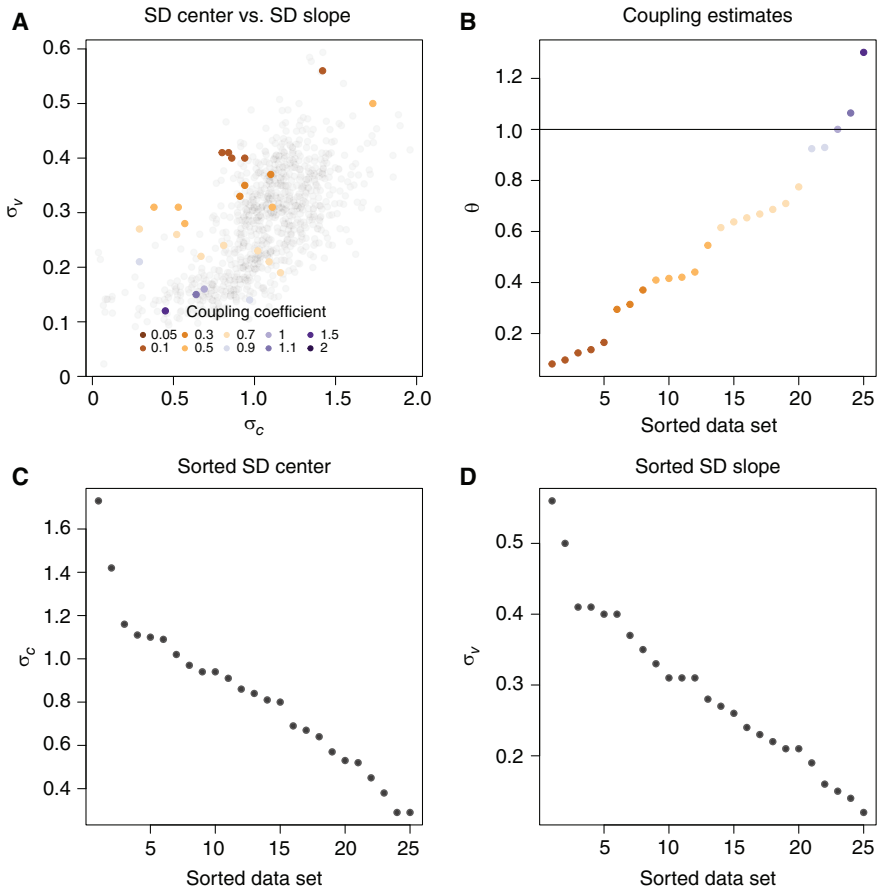


Figure 5. Cline variances and coupling in empirical hybrid zones. Panel *A* shows estimates of the standard deviation (SD) for genomic cline center (σ_c) and slope (σ_v) from the simulated (light gray points) and empirical (colored points) hybrid zones. Coupling coefficients for the 25 empirical hybrid zones were estimated from a linear regression model fit with the simulated hybrid zone data. Panel *B* plots the estimated coupling coefficients (θ) for the empirical hybrid zones sorted from smallest to largest. The horizontal line denotes $\theta = 1$. Panels *C* and *D* show estimates of the SD for genomic cline center (σ_c) and slope (σ_v) for each of the 25 empirical hybrid zone data sets, here sorted from the largest (high variability among clines) to smallest (low variability among clines). These data suggest a continuum of estimated coupling coefficients (*B*) and cline parameters SDs (*C*,*D*).

no evidence of a tipping point or positive feedback loop near $\theta = 1$ that would rapidly drive systems to higher levels of coupling resulting in a dearth of systems with $\theta \sim 1$ (analogous to the gap in genetic differentiation documented in sympatric *Timema* stick insects, see Riesch et al. 2017). Several factors likely contributed to our finding of a smooth continuum of inferred coupling coefficients. First, if species mostly diverge in allopatry with limited gene flow, there would be limited opportunity for the feedback of increased coupling to reduce effective migration and further increase LD among barrier loci. This is consistent with findings from theoretical work by Flaxman et al. (2014) that found evidence of a sudden transition to genome-wide congealing (analogous to coupling) when gene flow was strong relative to selection, but more gradual divergence otherwise (also see Sinitambirivoutin et al. 2023). However, this feedback could still operate upon secondary contact and thus in hybrid zones if species come into contact

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before speciation is complete (see, e.g., Flaxman et al. 2014). Second, different organisms vary in ecology and demographic histories, including the time since secondary contact, and these differences could act to smooth out the empirical distributions of cline SDs and estimates of coupling across systems (this is part of a general difficulty in treating pairs of species as a reconstructed "chrono-sequence"; see, e.g., Nosil et al. 2017; Bolnick et al. 2023). Third, different sets of loci, such as barrier loci versus neutral loci, can become coupled at different rates (Barton 1983; Barton and Bengtsson 1986; Schilling et al. 2018). Our analyses necessarily average over such variation, which could contribute to the continuum of coupling documented here. And finally, some predictions for a sudden transition in coupling in terms of parameter space may not reflect the temporal dynamics by which reproductive isolation evolves. For example, we may not see the expected transition from uncoupled to coupled dynamics at $\theta = 1$ when holding R and total selection constant but increasing L (Barton 1983) if speciation does not progress with total selection held constant.

Despite the smooth continuum we detected in terms of cline SDs and estimated coupling coefficients, we did observe patterns that suggest a transition in hybrid zone dynamics around $\theta = 1$. Specifically, in simulations, hybrids become increasingly rare around $\theta = 1$ and few of our empirical hybrid zones had estimates of θ notably larger than 1. Thus, consistent with past work (e.g., Barton 1983), our results suggest that by $\theta = 1$ the overall barrier to gene flow across much of the genome is quite strong. The small number of empirical hybrid zones with θ larger than 1 suggests that species pairs with such high levels of coupling rarely have patterns that evolutionary biologists would classify as hybrid zones. In other words, $\theta = 1$ might roughly approximate a genome-wide species boundary even if the approach to this boundary occurs gradually and at a near constant rate with increases in θ (e.g., with an increase in the number of barrier loci and thus a decrease in r).

Our results are relevant for at least two other classic issues in the study of hybrid zones and speciation. First, considerable work has been done on the consistency or variability of overall and locus-specific patterns of introgression across replicate hybrid zones or hybrid zone transects (Buerkle and Rieseberg 2001; Nolte et al. 2009; Janoušek et al. 2012; Larson et al. 2014; Schaefer et al. 2016; Mandeville et al. 2017). This is relevant as it bears on the degree to which reproductive isolation and hybridization outcomes are contingent on the ecological context of secondary contact. Such analyses have rarely, if ever, considered cline SDs and coupling explicitly when comparing hybrid zones. We found relatively high levels of consistency even in systems where locus-specific patterns of introgression have been shown to be less consistent, that is, where patterns of introgression for individual loci have varied substantially across transects (e.g., *Mus*; Teeter et al. 2010). Thus, our results hint at a greater consistency in the overall barrier to gene flow, as captured by cline SDs and estimates of θ, than in locus-specific patterns. Second, there is considerable evidence that sex chromosomes (X or Z) contribute disproportionately to reproductive isolation and exhibit reduced introgression in hybrid zones (Tucker et al. 1992; Carling and Brumfield 2008; Janoušek et al. 2012; Hooper et al. 2019). Our results suggest that, consistent with these other patterns, sex chromosomes show higher levels of coupling (lower cline SDs) than autosomes, which could be at least partially responsible for the lower overall rates of introgression (Muirhead and Presgraves 2016).

Our simulations and analyses documented a link between genome-wide variation in clines and a theoretical quantity relevant for understanding the extent to which genes or the entire genome experience a barrier to gene flow (the coupling coefficient, θ). Using simulations, we showed that this relationship is somewhat consistent under different conditions, but does vary to an extent and is likely to vary even more under conditions that diverge more from those we considered, such as in mosaic hybrid zones lacking a strong spatial axis. Our focus here was on cline SDs, but additional information, such as the prevalence of hybrids or the distribution of hybrids and introgression in space, could provide additional information about coupling. Using

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such additional metrics perhaps combined with more tailored simulations based on individual hybrid zones could provide a powerful framework to infer coupling coefficients using, for example, approximate Bayesian computation or neural networks.

More generally, our results demonstrate the relevance of quantifying cline variances rather than focusing solely on patterns of introgression for individual loci in analyses of hybrid zones. Under some conditions, especially strong coupling in the later stages of speciation, cline variances could be more informative about the process of speciation than patterns of introgression for individual loci. In contrast, when coupling is weak, loci resistant to introgression would be more likely to reside in genomic regions causally connected to reproductive isolation (Gompert et al. 2012). Additional understanding could come from quantifying cline variances and coupling at different genomic scales. For example, here we documented differences in variances and coupling for autosomes versus sex chromosomes. Finer scale analyses looking at coupling along chromosomes (e.g., in megabase windows) or within versus outside of large structural variants could provide additional insights on how the ratio of selection to recombination varies across the genome and thus on the genetics of speciation. In light of our findings, we think further empirical analyses of cline variances and coupling focused on the transition between weak and strong coupling, ideally within specific taxonomic groups, are critical for advancing understanding of the dynamics of speciation.

DATA AND CODE AVAILABILITY

Simulated hybrid zone data sets and our formatted input files for the empirical hybrid zones are available on Zenodo (doi: 10.5281/zenodo.8231089). Computer scripts used for simulations, data processing, and analyses are available from GitHub, github.com/zgompert/ClineCoupling.

AUTHOR CONTRIBUTIONS

All authors conceived of and designed the study. T.J.F. compiled the hybrid zone data, with help from G.S., S.A.T., E.L.L., and Z.G. Z.G. conducted the hybrid zone simulations and analyzed the data. T.J.F. and Z.G. wrote an initial draft. All authors contributed substantially to editing and revising the material.

ACKNOWLEDGMENTS

The support and resources from the Center for High Performance Computing at the University of Utah are gratefully acknowledged. The National Science Foundation provided support for this project to Z.G. (NSF DEB 1844941), T.J.F. (NSF DBI 2208825), E.L.L. (NSF DEB 2012041), and S.A.T. (NSF IOS-1754909; DEB-192889). Silhouettes for Figure 4 were obtained from phylopic.org. We thank our colleagues who generated each of the 25 empirical hybrid zone data sets for making these data publicly available or providing them to us upon our request.

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